LiveSearchVQA: Auditable Construction and Diagnosis of Renewable Multimodal Search

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Abstract

Live multimodal search requires agents to resolve visual references, obtain recent evidence, and select the fact that answers a question. A final-answer score conflates these operations, while recent publication alone establishes neither search need nor answerability. We propose LiveSearchVQA, a renewable benchmark protocol that attaches an auditable admission record to each item. Candidates undergo visual-grounding checks and repeated tests in a declared model panel: every closed-book trial must fail and every gold-evidence trial must succeed. This certificate is finite and panel-relative. An exact short-circuit implementation avoids unnecessary trials on rejected candidates, and an independent expert audit assesses validity beyond model agreement. Evaluation combines closed-book, with-search, and gold-evidence conditions with a joint analysis of evidence availability and correctness. Six reproducible numerical studies exercise audit design, paired scores, joint diagnosis, context interventions, visual controls, and construction. They include overlapping performance profiles, reranking losses, incomplete held-out transfer, and daily release shortfalls. These simulations verify accounting and expose testable alternatives; they do not establish performance or failure causes in real agents. The resulting protocol links construction, quality assurance, and diagnosis through versioned item records and paired evaluation.

1 Introduction

A question about a pictured product's newly announced price is not answered by object recognition alone. The agent must identify the product, locate the relevant announcement, and distinguish the requested price from a list price or an older promotion. Conversely, a question asking only what the picture shows may be answered without search. Both can appear in a fresh-news collection, but they measure different capabilities.

Freshness also does not resolve why an agent fails. A retrieved page can contain the answer without the relevant passage reaching the model's context. An agent can see the supporting fact yet choose a nearby number. Figure 1 separates these possibilities. Treating every wrong answer as a retrieval miss, or every correct answer as evidence of successful search, obscures what the evaluation measures.

We address this measurement problem by connecting construction controls to evaluation records. An evidence-first generator proposes image-grounded, event-specific questions. A separate verifier retains the visual audits and every trial underlying a joint closed-book/gold-evidence admission decision. An independent expert audit checks whether the resulting questions and labels are valid. Each evaluated agent then receives the same frozen items in three fresh contexts, with evidence availability recorded from the actual tool content shown to the agent.

Our contribution is the *joint, auditable protocol*, rather than an individual filter or diagnostic metric. Earlier work already screens no-search answerability, validates source-conditioned answers, and separates retrieval from generation. We integrate these controls with renewable visual questions, finite repeated-trial certificates, and explicit release versions. We also give an exact early-stopping rule and a reproducible numerical demonstration of the paired analysis. Whether this combination

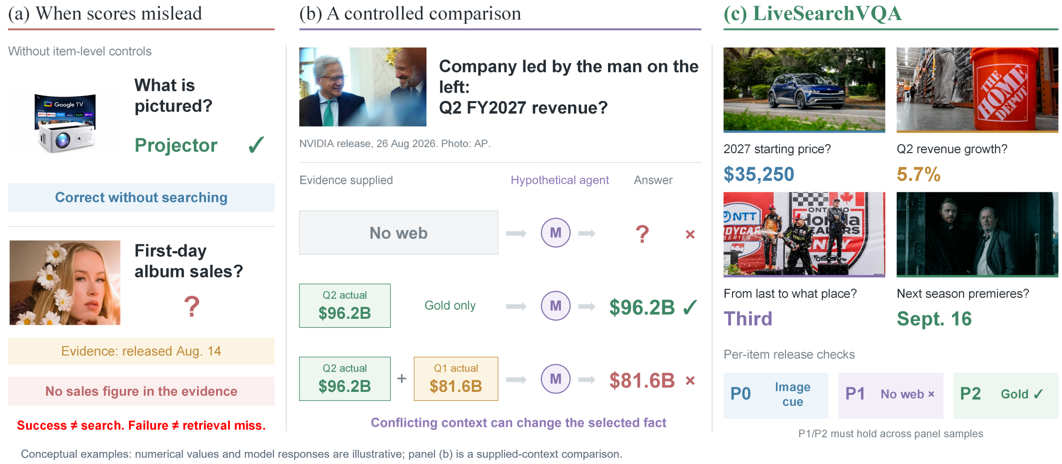

Figure 1: **Why final accuracy needs item and context controls.** (a) A recognition question may not require search, whereas a recent source can omit the requested fact. (b) A conceptual comparison of no evidence, gold evidence, and gold plus conflicting context motivates a controlled evidence-use test; the third row is not a live-search run. (c) Example task forms motivate visual, closed-book, and evidence-conditioned checks. Numerical values and model responses are illustrative. Released questions must specify their event and time, beyond the shorthand shown here.

improves real benchmark validity is an empirical question addressed by the expert, cross-panel, and intervention studies specified below.

2 Related Work and Positioning

Prior work establishes several components of our protocol. Dyn-VQA uses visual co-references to replace named entities (Li et al., 2024), and LiveVQA retains contextual questions only when a model answers them correctly from the image and source context (Fu et al., 2025). MMSearch-Plus provides a direct P1 antecedent: its *lite* subset removes every question answered without search by any tested model (Tao et al., 2026). LiveNewsBench combines refreshed textual news questions with source-based verification (Zhang et al., 2026). Diagnostics also have established precedents: RAGChecker separates retrieval coverage, context utilization, and noise sensitivity (Ru et al., 2024), while SealQA probes reasoning with conflicting retrieved documents (Pham et al., 2025). We study the integration of these controls into an auditable contract for renewable visual questions: a declared panel, repeated closed-book and evidence-conditioned trials, and retained item-level records link construction decisions to subsequent diagnostics.

MMSearch evaluates requery, reranking, summarization, and end-to-end search; VDR-Bench combines answer accuracy with entity recall and checks textual shortcuts (Jiang et al., 2024; Zeng et al., 2026). Thus, neither stage-level diagnosis nor visual-dependence screening is absent from existing benchmarks. Table 1 compares concrete construction and evaluation choices. The distinction we aim to test is whether a refreshed item contract that retains both screening outcomes and subsequent evidence traces makes measurement errors easier to detect and reproduce. This does not imply that our protocol dominates tasks designed for longer search horizons or different forms of visual reasoning.

3 Auditable Benchmark Construction

3.1 Renewal and the item contract

The reference release schedule targets 200 English-news items per day over 30 days. This is a design target, not a quota that relaxes admission rules: a shortfall is released and reported as a shortfall. Figure 2 shows the workflow. Each item stores

$$x = (I, q, a^*, e^*, u, t_{\text{pub}}, t_{\text{build}}, v), \tag{1}$$

Benchmark	Data and renewal	Documented construction controls	Evaluation emphasis
MMSearch (Jiang et al., 2024)	Image–text search; 300 curated tasks.	Human curation of multimodal search questions.	Requery, rerank, summarization, and end-to-end scores.
Dyn-VQA (Li et al., 2024)	Bilingual VQA; changing answers with scheduled updates.	Visual co-reference rewriting and human quality checks.	Adaptive retrieval and multi-hop reasoning.
LiveVQA (Fu et al., 2025)	Recent news, videos, and papers; collected release.	Image relevance checks; source-context answer filtering (P2 antecedent).	No-search/search comparisons and knowledge updating.
MMSearch-Plus (Tao et al., 2026)	Image–text browsing; 311 curated tasks.	**Lite subset:** excludes items solved without search by any tested model (P1 antecedent).	Provenance, crop/search tools, and error analysis.
LiveNewsBench (Zhang et al., 2026)	**Textual** news QA; regularly refreshed.	Source-cluster self-consistency; automated and human validation.	No-search/search and search-budget comparisons.
VDR-Bench (Zeng et al., 2026)	Visual multi-hop QA; 2,000 curated tasks.	Recorded-path solvability and human checks for textual or prior-based shortcuts.	Answer accuracy and entity recall.
Our protocol	Image-grounded news QA; on-demand builds within 48 hours.	P0 plus declared-panel P1/P2; repeated trials and their records retained per item.	Three conditions, trace diagnostics, and controlled noise probes.

Table 1: **Positioning through documented design choices.** P1 and P2 have antecedents in prior item filtering and source-context verification. Our row specifies the proposed joint admission and diagnostic protocol; the contribution is its integration and auditable record.

where I is the image, q the question, a^* the answer, e^* a supporting source span, u the source URL, and v the immutable item version. The publication window is checked at release: $0 \leq t_{\text{build}} - t_{\text{pub}} \leq 48$ hours. The record also includes archived source content, image provenance, event identifiers, evidence offsets, and the full verification record. Event dates and publication dates are distinct fields. A question about a changing value names its event or reference time; "current" is never silently reinterpreted at a later evaluation date.

3.2 Evidence-first proposals and finite certificates

The generator first selects a recent, event-specific fact and its exact supporting span, then writes a question whose referent is resolved through the image. Its rejection memory stores typed errors and accepted examples for later builds. These updates are versioned and frozen during each build. A same-call self-check can reject weak proposals, but it has already seen the evidence: it is an evidence-aware heuristic, not a closed-book test.

P0: audited visual grounding. The verifier checks image–source alignment, identifiable visual reference, event specificity, and whether pixels alone expose the requested answer. The reference rubric uses an image–source match score of $4/4$, a question–image grounding score of at least $2/4$, and two consistent referent judgments; Appendix B defines the rubric. These gates establish recorded audit decisions. Image necessity requires additional text-only and matched image-replacement tests; omitting an entity name does not by itself establish necessity.

P1/P2: repeated panel tests. Fix a panel $\mathcal{M}$, R trials per model, and an answer-equivalence rule $\mathrm{Eq}(\hat{a}, a^*; q, e^*)$. Write $C_{mr}(x)$ and $O_{mr}(x)$ for the correctness of a fresh closed-book response and a fresh gold-evidence response, respectively. Admission requires

$$\mathrm{Accept}(x) = P0(x) \;\wedge\; \underbrace{\bigwedge_{m\in\mathcal{M}, r\leq R} \neg C_{mr}(x)}_{P1(x)} \;\wedge\; \underbrace{\bigwedge_{m\in\mathcal{M}, r\leq R} O_{mr}(x)}_{P2(x)}\,. \tag{2}$$

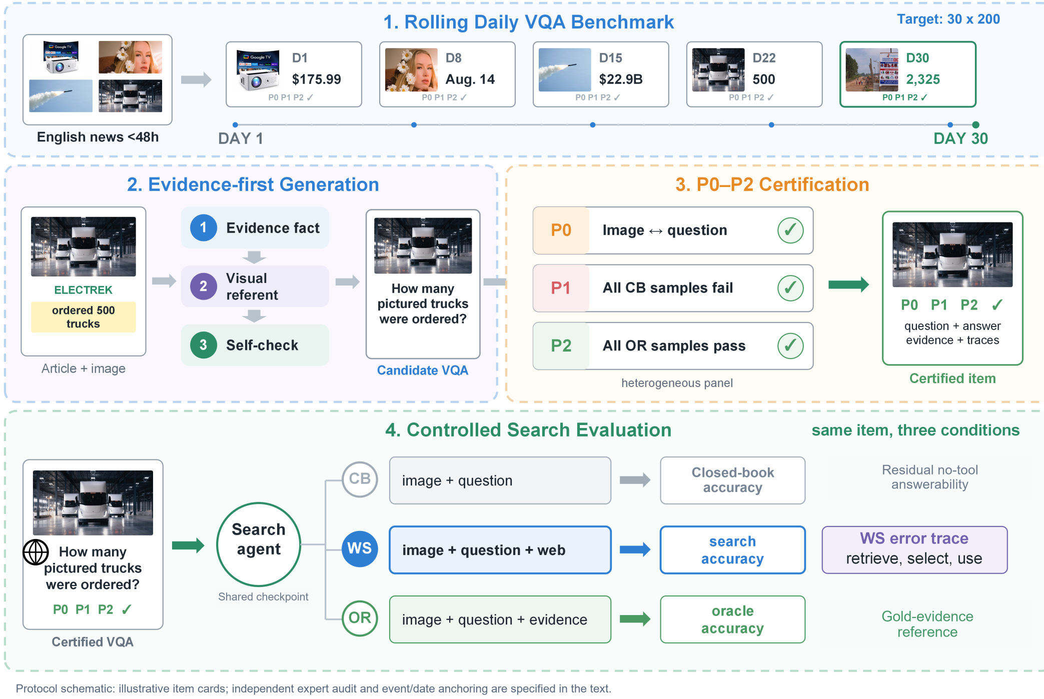

Figure 2: **Renewable construction with auditable admission and evaluation.** Recent sources feed evidence-first proposals. P0 audits visual grounding; P1/P2 require all recorded closed-book failures and gold-evidence successes in the declared panel. Evaluation uses separate contexts for the same checkpoint in CB, WS, and OR conditions. The 30-by-200 schedule is a target; item cards abbreviate event-anchored questions. A sampled independent expert audit complements automatic admission (Appendix A). CB and OR are measured reference conditions, not guaranteed bounds.

The reference profile uses three models and four trials, hence 12 CB and 12 OR responses for each automatically admitted item. CB receives only I, q and the common answer instruction; source text, answers, filenames, URLs, and previous outputs are excluded. OR adds the specified evidence. Grading happens outside the respondent context. Every trial's input, output, model version, decoding configuration, and verdict is retained.

The certificate means *no observed CB success and no observed OR failure in these trials*. It does not prove universal search necessity, zero future CB accuracy, or source truth. Shared model errors can satisfy P2, motivating independent expert validation. Reusing the same panel predictions as test outcomes would make CB/OR endpoints tautological; evaluation uses new runs and reports construction-panel membership separately.

3.3 Efficiency and independent quality assurance

CB rejects on its first correct answer; OR rejects on its first incorrect answer. For fixed candidates, P0 decisions, trial outputs, grading, and release postprocessing, this short-circuit rule produces exactly the same accepted set as exhaustive evaluation. Every accepted item still has all 24 responses. The proof and cost baselines appear in Appendix C. Changing the generator or a filtering threshold changes the candidate distribution and is not covered by this equivalence.

Independent expert audit. The protocol requires a probability sample of frozen items to be reviewed independently of construction. Qualified reviewers first inspect image and question without the answer, then derive an answer from the archived source before seeing the proposed gold label. They assess correctness, event unambiguity, visual reference, pixel-only shortcuts, evidence sufficiency, and temporal validity. They are blind to generator identities, panel verdicts, and evaluated-agent outcomes. Independent labels and adjudication are retained separately. A sampled audit estimates release quality; it does not certify every item. Appendix A gives the complete protocol, reporting fields, and an analytical sample-size study. Human judgments do not substitute for the model-specific P1 tests.

4 Diagnostic Evaluation

4.1 Three reference conditions

For every frozen item, CB supplies image and question; WS additionally allows the agent to obtain evidence with the declared tools; OR directly supplies the gold span without retrieval. The checkpoint and common answer instruction are shared, and each condition starts in a new context. WS is evaluated under a fixed search endpoint, tool-call budget, context limit, and time window. Exact settings and failures belong to the release record (Appendix D).

Report $\mathrm{Acc}_{\mathrm{CB}}$, $\mathrm{Acc}_{\mathrm{WS}}$, and $\mathrm{Acc}_{\mathrm{OR}}$ first. The optional normalized difference

$$\eta = \frac{\mathrm{Acc}_{\mathrm{WS}} - \mathrm{Acc}_{\mathrm{CB}}}{\mathrm{Acc}_{\mathrm{OR}} - \mathrm{Acc}_{\mathrm{CB}}} \tag{3}$$

describes improvement relative to the measured CB–OR gap when the denominator is positive. It is undefined when that gap is zero and should not be treated as a fraction of available capability when the denominator is negative. There is no general $[0, 1]$ bound: WS may exceed OR or underperform CB. Neither CB correctness nor a CB–WS difference identifies memorization alone.

4.2 Evidence availability and conditional correctness

Let $C = 1$ indicate a correct WS answer and $G = 1$ indicate that sufficient support for the requested fact was present in the context actually shown to the agent before its final answer. Validated equivalent sources count; a URL hit without the fact, an unopened page, or a truncated span does not. For fully resolved availability labels, define

$$\mathrm{Cov} = \Pr(G = 1), \qquad U_1 = \Pr(C = 1 \mid G = 1), \qquad U_0 = \Pr(C = 1 \mid G = 0). \tag{4}$$

Then the exact accounting identity is

$$\mathrm{Acc}_{\mathrm{WS}} = \mathrm{Cov}U_1 + (1 - \mathrm{Cov})U_0. \tag{5}$$

Correct answers without observed supporting evidence remain in U_0; they may arise from prior knowledge, guessing, or an availability-label error. They must not be included in the numerator of U_1. In particular, $\mathrm{Acc}_{\mathrm{WS}}/\mathrm{Cov}$ is not generally conditional evidence-use accuracy. The four joint cell counts, not only marginal percentages, are released.

For errors, the primary categories are *support absent*, *support present*, and *availability unresolved*. Stale-source matches, contradiction of visible evidence, and competing-number selection are additional, potentially overlapping tags. OR failure is a separate observation, not a cause assigned to a WS error. Likewise, observing a distractor does not establish that it caused the answer. Unresolved availability remains explicit; the corresponding three-way identity and coverage bounds are given in Appendix D.

4.3 Interventions and validity tests

Supplied-context tests compare clean gold evidence, gold plus irrelevant passages, and gold plus conflicting passages. The latter two match length, number, source cues, and gold position; the clean-to-irrelevant contrast measures context burden, while conflict-to-irrelevant isolates the chosen conflict treatment. Retrieval-depth and reranking tests additionally report coverage, since reranking can discard the relevant fact. Image removal, matched image substitution, and a text-only entity-name control test the contribution of the visual reference. Fresh runs from a family-disjoint panel test certificate transfer. These are separate tests: success on one does not establish the others.

5 Experiments: Six Tests of the Evaluation Protocol

Design and statistical unit. The numerical study exercises six complementary parts of the protocol. An evaluation fixture contains 30 builds of 200 items, grouped into 20 events per build, and 12

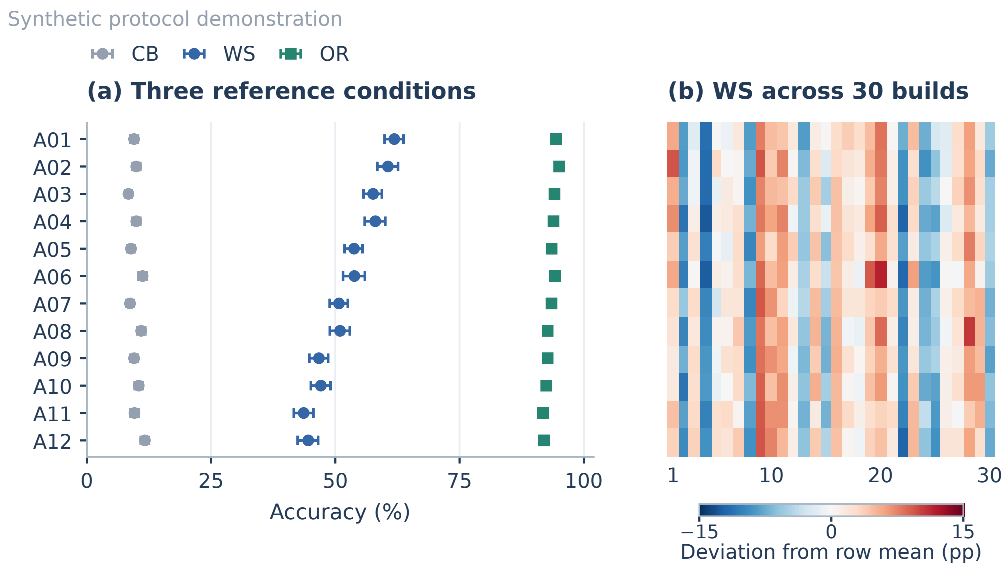

Figure 3: **Three conditions and variability across builds.** Synthetic results for 12 fixed configurations on the same 6,000 items. (a) CB, WS, and OR with paired-build marginal intervals. (b) Each row shows the 30 build scores after subtracting that configuration's mean WS score; the color scale is shared. The fixed row order is identical in subsequent figures. Full estimates and paired contrasts appear in Appendix G.

anonymous configurations A01–A12. A separate construction stream contains 18,000 candidates. Neither contains real model runs or expert labels; the audit study is analytical. Fixed seeds and assumptions are documented before generating each extension. Evaluation outcomes are generated once, without score fitting or seed search. Unless stated otherwise, intervals are pointwise 95% percentile intervals from 2,000 paired resamples of whole builds, recomputing all conditional rates from counts. They quantify variation under the synthetic design. Appendices F–I provide equations, complete results, and independent record-based checks.

5.1 Experiment 1: Can the quality audit detect defects?

Automatic agreement is insufficient to establish source correctness or visual validity. The independent expert protocol in Appendix A therefore uses three reviewers, a two-stage blind interface, six criteria, and separate pre-adjudication labels. Its analytical sample-size study distinguishes sampling coverage from recognition. For a fixed pool of 6,000 items with 60 defects, uniformly auditing 300 includes at least one defect with probability 95.5%. If a sampled defect is independently recognized with probability $s = 0.8$, detection falls to 91.4%; 364 items are needed to reach 95% under that assumption (Figure 7). At $s = 0.6$, the corresponding sample size is 486. These are planning calculations, not observed reviewer sensitivities. They show why release validity, grader calibration, and evidence-availability annotation each need an independent reference check.

5.2 Experiment 2: What do the three conditions reveal?

Figure 3 separates end-to-end scores from the no-evidence and supplied-evidence references. WS ranges from 43.6% to 61.9%, while OR ranges from 91.78% to 95.05%. Configurations have different retrieval and reading parameters, and the shared build terms produce correlated score changes over time. This distinguishes variability within a fixed fixture from changes in the evaluated system.

Close averages require paired analysis, not visual inspection of marginal interval overlap. A03 minus A04 has a WS difference of -0.45 percentage points (pp), with interval $[-1.18, 0.28]$. A05 minus A06 is -0.07 pp $[-1.12, 0.92]$. The full adjacent-pair table retains both gains and losses on individual items; these descriptive contrasts do not establish a total ranking or support multiple unadjusted superiority claims.

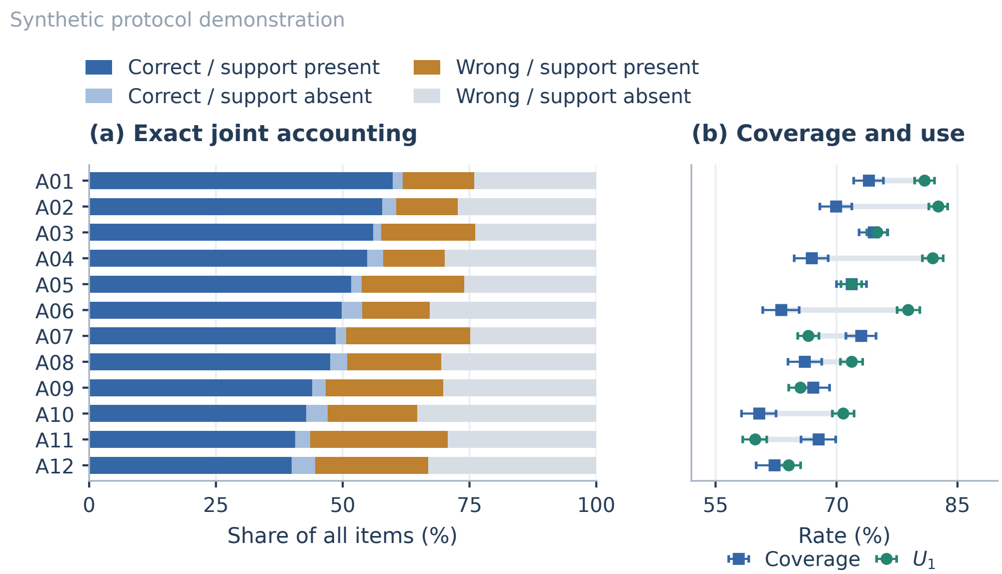

Figure 4: **Evidence availability and correctness answer different questions.** (a) Four disjoint cells partition every configuration's 6,000 synthetic records. (b) Coverage and conditional correctness U_1 use different denominators; connecting them is a visual comparison, not a trajectory. Intervals recompute the relevant ratios in each build resample. The cells describe observed contexts and do not attribute a causal error mechanism.

5.3 Experiment 3: Where do incorrect answers occur?

Figure 4 reports the joint distribution rather than assigning every WS error to retrieval. A03 observes support more frequently than A04 (74.6% versus 66.9%), but has lower conditional correctness (75.1% versus 81.9%). Thus a coverage ranking need not match a reading ranking. Correctness without observed support remains nonzero: U_0 ranges from 6.3% to 12.3%. Removing that group would inflate estimated utilization and break Equation 5. All 48 joint counts and their denominators are released. An independent CSV computation reproduces the exact decomposition to floating-point precision. Real traces additionally require the unresolved category and matcher-error audit defined in Appendix D.

5.4 Experiment 4: Which interventions help, and at what cost?

In the four selected configurations, irrelevant context changes OR by -0.2 to -1.3 pp. Conflicting context has an additional -6.9 to -14.0 pp effect relative to the matched irrelevant control (Figure 5a). The latter contrast isolates the specified conflict perturbation from generic added-context burden.

Deeper retrieval does not imply better answers. Increasing depth from 3 to 20 improves WS by 11.0 pp in A01 and 11.8 pp in A04, but reduces it by 1.4 pp in A07 despite higher coverage. Reranking instead changes WS by -1.7, -2.0, and $+7.5$ pp in those configurations; A10 improves by 4.1 pp. For A01, reranking corrects 80 previous errors but introduces 183, making its net loss auditable. These effects follow the stated numerical sensitivities. Their empirical counterparts require a gold-blind reranker, matched retrieval snapshots, and measured context exposure, as specified in Appendix E.

5.5 Experiment 5: Does a visual control establish necessity?

The visual-control extension reuses every full-image outcome and compares image removal, a within-build donor from a different event, and a text-only condition that names the referent without supplying its answer. The shared uniforms and fixed perturbation equations retain pairing across all 12

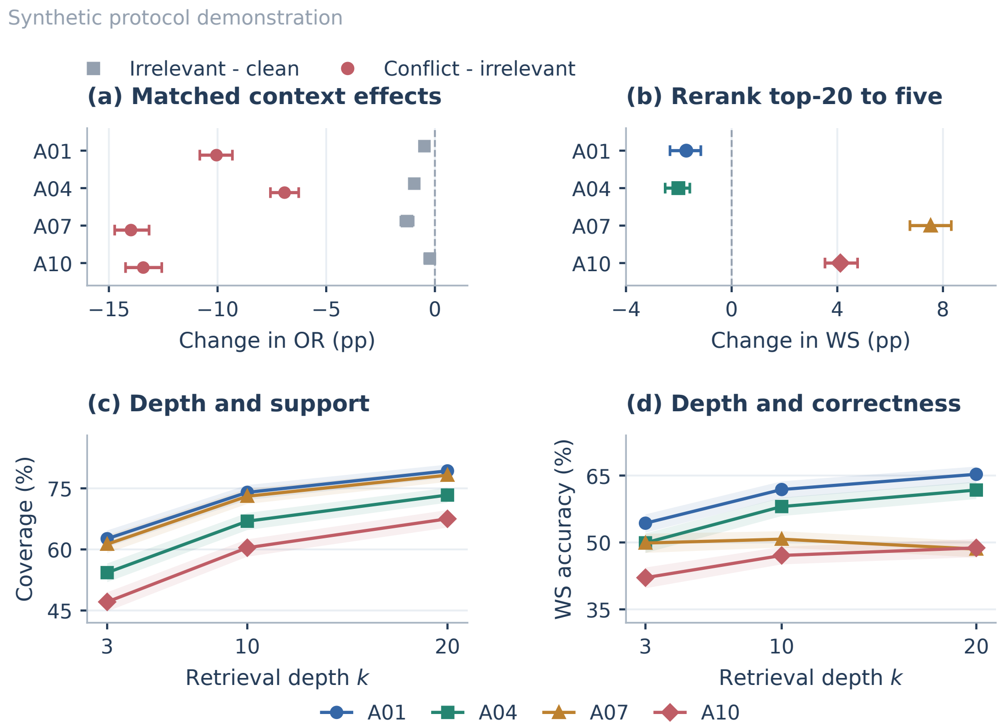

Figure 5: **Paired interventions with both benefits and losses.** (a) Matched context contrasts separate burden from conflict. (b) Reranking compares top-20 context against five retained units. (c,d) Depth changes evidence coverage and WS correctness separately. The four configurations and all conditions were fixed before generation. Bars and shading show pointwise paired-build intervals; context units are numerical proxies, not retrieved passages. Complete paired transitions are retained with the raw records.

configurations. A familiarity-based shortcut proxy defines two numerical strata; it is not an annotation of actual question shortcuts.

Across configurations, removal changes WS by -2.09 pp $[-2.24, -1.93]$, shuffling by -3.20 pp $[-3.36, -3.04]$, and naming the referent by $+1.58$ pp $[1.43, 1.76]$ (Figure 9). Removal nevertheless retains 95.80% of originally correct configuration–item outputs. Its effect ranges from -0.63 pp in A04 to -2.97 pp in A09, following the fixed sensitivities. Thus a mean omission drop alone is insufficient to establish necessity. Appendix H reports all configurations, proxy strata, and paired gains and losses. Actual images, controlled prompts, and expert review are needed to test the analogous claim on real questions.

5.6 Experiment 6: Is construction efficient and transferable?

The candidate stream caches every trial before applying five admission policies to the same P0-eligible pool. Of 18,000 candidates, 15,459 pass P0 and 7,004 pass full P1/P2. Only 4,079 of these (58.24%, interval $[56.21, 60.23]$) also pass both predicates in the held-out synthetic panel (Figure 6a). Family labels are disjoint across panels, while candidate and build factors remain shared. This illustrates selection and finite certificate scope; the policies select different cohorts and do not estimate fixed-set performance gains.

Exhaustive, block-skipping, and exact short-circuit execution require 371,016, 319,656, and 265,196 checked predictions, respectively. Short-circuit savings are 28.52% $[26.31, 30.69]$ relative to exhaustive evaluation and 17.04% $[15.87, 18.30]$ relative to block skipping. Independent replay confirms identical admitted and released IDs and all 24 checked predictions for each admitted item. This is an execution result for fixed inputs, not a claim about API bills or generator adaptation.

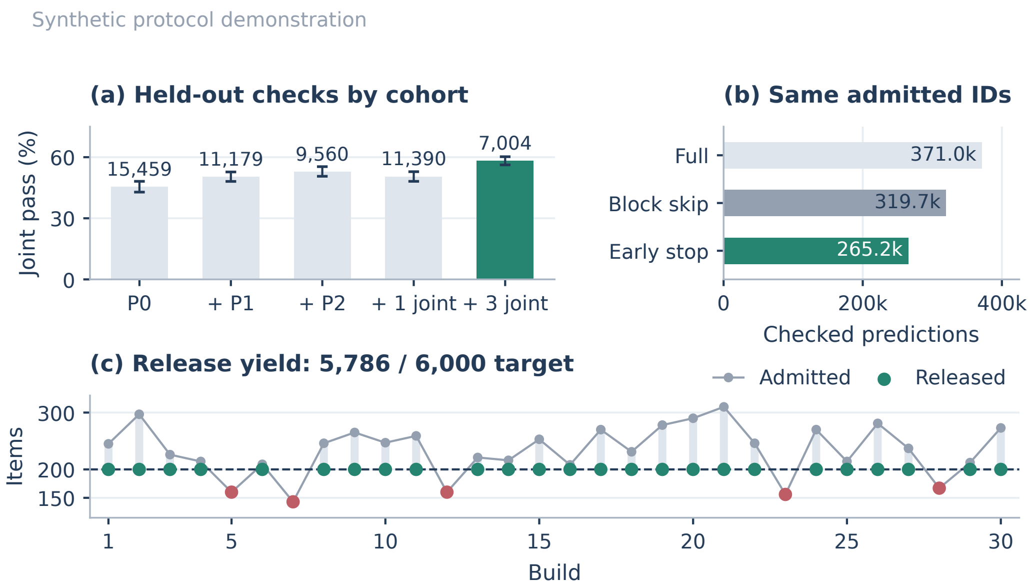

Figure 6: **Selection, execution, and yield in a separate construction stream.** (a) Held-out joint pass rates on each policy's own admitted cohort; labels give cohort sizes. Each added filter starts from the same P0 pool. (b) Three execution procedures preserve the same admission and release IDs. (c) A 200-item cap leaves five builds below target, highlighted in red. All outcomes are synthetic; prediction counts assume equal unit costs. Appendix I gives both admitted and released joint tables.

The daily cap releases 5,786 items; five builds leave a total shortfall of 214. Surplus admissions in other builds do not fill those missing slots. The released cohort has its own joint transfer rate, 57.38% $[55.35, 59.38]$. These 5,786 selected items are distinct from the independently generated 6,000-item evaluation fixture used in Experiments 2–5.

6 Limitations

Certification selects for the tested panel and decoding settings; repeated errors can be correlated. News recency does not guarantee contamination freedom. The English-news window restricts task coverage, while strict P2 may favor short evidence over complex search questions. P0 and trace labels can be wrong, and expert sampling does not validate every item. Sources can themselves be incorrect. Web/model updates limit replay, requiring separate live and archived evaluations. Source and image provenance and redistribution permissions must accompany a release.

The generator is not fitted to browsing logs. Its context effects and correlations are assumptions; intervals concern the fixture alone. It omits actual vision, language ambiguity, and tool failures. The additional construction stream exercises selection under assumed probabilities rather than estimating the selection distribution of news questions. Empirical deployment, expert annotation, and common-scaffold comparisons are needed to establish benchmark quality or superiority.

7 Conclusion

LiveSearchVQA specifies a renewable visual-search evaluation in which each item links source evidence, finite admission trials, and subsequent diagnostic records. Exact short-circuit certification preserves the declared admission decision, while expert review addresses validity beyond model agreement. The paired numerical demonstration shows how to report evidence acquisition, conditional correctness, and intervention tradeoffs without conflating them. The protocol makes its empirical claims directly testable through released item-level records.

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A Independent Expert Audit

This appendix specifies an audit to be executed on a frozen empirical release. It does not report completed expert annotation. Automatic panel agreement is not the human reference standard, and all simulated performance records in Section 5 are outside this audit's sampling frame.

A.1 Sampling, expertise, and independence

The reference design samples 300 items uniformly without replacement from a 6,000-item release, with the actual release size substituted when daily production falls short. The target population is release *items*; claims about unique events require a separate event-weighted analysis. Supplemental samples of rare answer types, rejected candidates, or disputed items are reported separately with known inclusion probabilities. They are not mixed into an unweighted release-quality estimate.

Three independent reviewers examine each sampled item. Reviewer recruitment requires documented competence in English source interpretation and the relevant domains, including numeric, financial-period, and date conventions where applicable. A specialist can adjudicate an unfamiliar domain. The report records actual qualifications, training, compensation, conflicts, and any role in pipeline development; job titles alone do not establish expertise. Reviewers must not have written the items they audit. A separate calibration set is used to refine the rubric before the sample is drawn. Its labels are excluded from the final quality estimates.

The sampling seed, item versions, source snapshots, and rubric are frozen before inspection of evaluated-agent results. Reviewer interfaces hide generator names, certification verdicts, model identities, and model performance. Human review addresses whether a question is valid and its answer supported. Inability of a reviewer to answer without a source does not establish that a model must search.

A.2 Two-stage annotation and adjudication

Stage 1 displays only the image, the complete question, and any time anchor included in the agent input. Reviewers identify the visual referent, check its uniqueness, mark a directly visible answer, and flag textual shortcuts or unresolved event references. Stage 2 displays the archived source with publication and event metadata. Reviewers independently extract the supported answer *before* the proposed gold label is revealed. They then compare the answer and the minimal supporting span against the proposed record.

Criterion	Required evidence for a pass
Answer correctness	The source supports the exact entity, value, unit, date, and requested relation. Faithful citation is distinguished from independent corroboration of the source's claim.
Event unambiguity	The question and its time anchor identify one relevant event; alternative orders, releases, quarters, or promotions do not fit equally well.
Visual reference	The supplied image supports identifying the intended referent and matches the source entity or event.
No pixel-only answer	Reading or recognizing the supplied pixels does not directly reveal the requested fresh fact. This is a shortcut audit, not a universal impossibility claim.
Span sufficiency	The supplied gold span, with exactly the metadata allowed in OR, supports the answer without an unstated additional source.
Temporal validity	Publication freshness satisfies the release rule and the answer refers to the specified event/time rather than a stale value.

Table 2: Expert audit rubric. Each criterion receives pass, fail, or unresolved with a cited rationale. Missing annotations are tracked separately.

Each annotation stores item version, reviewer identifier, criterion, pass/fail/unresolved verdict, source or image rationale, and annotation time. The initial labels are independent. A criterion majority requires at least two identical labels; a three-way split is unresolved. Consensus discussion cannot replace these original labels when computing agreement. Adjudication records the final verdict and supporting evidence separately.

A source error, ambiguous question, unsupported gold span, or invalid visual link triggers investigation and a versioned correction or withdrawal. The release report retains pre-correction quality, affected IDs, reasons, and the decision date. Revised items repeat the affected P0/P1/P2 steps and agent evaluation. Checking and then repairing a sample does not create an unbiased estimate of the repaired population; use an independent follow-up sample for that claim. A later price or event update is not automatically an error in a correctly time-anchored historical item.

A.3 Analytical sample-size study

For a fixed population of N items containing D defective items, a uniform sample of size n includes at least one defect with probability

$$P_{\text{detect}}(N, D, n) = 1 - \frac{\binom{N-D}{n}}{\binom{N}{n}}. \tag{6}$$

This finite-population calculation concerns sampling, not reviewer accuracy. It assumes a sampled defect will be recognized and does not require item defects to be independent: the population and its defective set are fixed. With imperfect reviewer sensitivity, actual detection can be lower. Table 3 is generated directly from this equation by `simulation/audit_sampling_design.py`.

Audited items	0.5% defects	1% defects	2% defects
100	39.7	63.7	87.0
200	63.9	87.1	98.4
300	78.6	95.5	99.8
500	92.7	99.5	$>$ 99.9

Table 3: **Analytical audit design, not observed annotation results.** Probability (%) of including at least one defective item when sampling uniformly without replacement from a fixed population of 6,000 items. Columns assume exactly 30, 60, or 120 defective items and perfect recognition of a sampled defect.

The 300-item design gives 95.5% inclusion probability at a 1% defect rate but only 78.6% at 0.5%. Observing no defects therefore cannot support an assertion of perfect quality. Observed quality estimates must report the audited denominator, pass/fail/unresolved counts, sampling weights where applicable, and intervals consistent with the actual design. Report each criterion separately and define a composite item failure in advance. Use finite-population intervals for the fixed release; inference to future news builds requires event/day-aware uncertainty.

Reliability is assessed from the full, pre-adjudication three-category label matrix, with raw category frequencies and a named multi-rater agreement statistic. Any binary analysis must disclose how unresolved labels were handled. High agreement does not establish correctness, and no reliability coefficient can be inferred from majority pass percentages alone.

A.4 Sensitivity to imperfect defect recognition

If K defects fall in the audit sample and each is independently recognized with assumed probability s, the probability of at least one detection is

$$P_{\text{detect}}(N, D, n; s) = 1 - \sum_{k=0}^{\min(D,n)} \frac{\binom{D}{k}\binom{N-D}{n-k}}{\binom{N}{n}} (1-s)^k. \tag{7}$$

Here s is the item-level probability that the audit process recognizes a sampled defect, not the accuracy of an individual reviewer. False positives are excluded in this design calculation. The independent-recognition assumption can fail when defects share a cause or a reviewer misconception; therefore the sensitivity curve is not a reliability estimate. The main-text sample calculation uses a 6,000-item reference pool. If an actual release is smaller, use its actual size and an explicitly chosen integer defective count. The 5,786-item capped construction stream is a separate numerical experiment, not the assumed population of this audit.

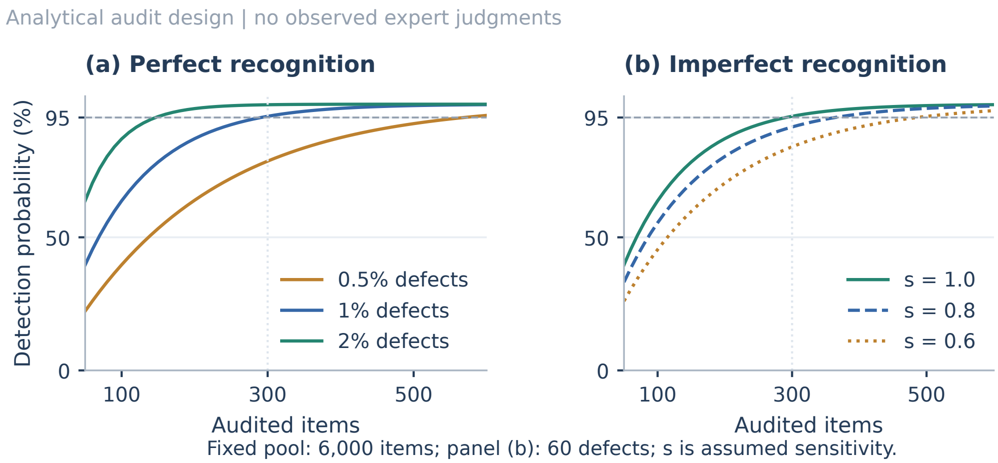

Figure 7: **Sample size and recognition are distinct audit assumptions.** (a) Exact detection for three fixed defect counts under perfect recognition. (b) Recognition sensitivity varies at a 1% defect rate. The horizontal line marks 95% detection and the vertical line the 300-item reference sample. No human labels or observed audit pass rates are represented.

Defects (%)	Assumed s	Detection at $n = 300$ (%)	n for 95% detection
0.5	0.6	59.95	950
0.5	0.8	70.68	712
0.5	1.0	78.62	569
1.0	0.6	84.01	486
1.0	0.8	91.45	364
1.0	1.0	95.46	291
2.0	0.6	97.47	246
2.0	0.8	99.28	184
2.0	1.0	99.80	147

Table 4: **Analytical sensitivity to imperfect recognition.** Fixed population 6,000; false-positive recognition is excluded and different sampled defects are recognized independently with probability s. The minimum integer sample size is obtained by exact hypergeometric summation. No sensitivity is estimated from humans.

A.5 Independent validation of scoring and trace labels

The answer grader receives the question, reference answer, and source span, but is blind to model identity. Its validation sample is stratified by agent, condition, predicted correctness, and answer type. Human reference labels are determined from source evidence independently of the automatic decision. The report includes weighted confusion counts, overall error, false-accept rate among human-incorrect responses, and false-reject rate among human-correct responses. Model-level error differences matter when assessing close rankings; an aggregate agreement percentage is insufficient.

A separate trace audit validates fact availability. Reviewers inspect the exact excerpts presented to the agent, not just search results or URL lists. They check equivalent sources, snippet/page truncation, tool errors, and whether the relevant content entered the context before the answer. Sample both matcher-positive and matcher-negative cases and retain an unresolved category. This audit distinguishes a matcher's limitations from an agent's retrieval behavior. The release includes the sampling frame, inclusion probabilities, raw labels, and adjudications for both studies.

B Item Record and Verification Rubrics

Each release includes the item version, image bytes or authorized reference, source URL and content hash, archived text, publication/build/evaluation timestamps, event ID, full question, answer type, canonical answer, accepted aliases, and exact supporting offsets. It also retains the generator and constitution versions, rejection stage, visual-audit explanations, panel model/API versions, prompts, decoding settings, response outputs, grader version, and trial verdicts. Actual data, unavailable artifacts, and access restrictions are explicitly enumerated in a release manifest.

Deterministic validity. The quality module checks the release-time freshness window, usable image, exact evidence occurrence, answer–span consistency, and uniqueness of the item ID. A numeric-token match is necessary for some extraction tasks but not sufficient: a price, percentage, date, and quantity can all occur in the same article. The question must specify the relevant relation. A source title or surrounding sentence is included in OR only if declared as part of the evidence input; otherwise a span requiring that hidden context fails.

P0 score definitions. Image–source alignment uses five ordered scores: 0, contradictory; 1, unrelated or merely generic; 2, plausible but ambiguous; 3, correct entity with unresolved event association; 4, verified entity/event correspondence at the level the question requires. The reference threshold is 4. Question–image grounding uses 0, irrelevant image; 1, weak or decorative association; 2, an identifiable referent; 3, clear disambiguation among plausible referents; 4, fine-grained visual cues essential to the intended resolution. The reference threshold is 2, with two separately prompted referent judgments required to agree. These are ordinal audit scores, not calibrated probabilities. Expert review and visual controls assess their external validity.

Strict P1/P2 profile. The design uses $|\mathcal{M}| = 3$, $R = 4$. CB uses temperature 0.7 and top-p 0.95 where supported; OR uses temperature 0.2 with the same top-p. The asymmetry is deliberate and recorded. Separate contexts are mandatory; “four trials” does not imply statistically independent errors. A deployment must record actual API capabilities and any unsupported controls. The source project lists `doubao-seed-2.0-pro`, `qwen3-vl-plus`, and `qwen3.5-flash` as a proposed panel, spanning two providers. These configuration strings do not establish current availability or completed runs. Evaluated checkpoints and family overlap must be documented from the actual run, and the anonymous simulation is not mapped onto these labels.

Answer equivalence. Normalize Unicode, whitespace, currency symbols, and declared unit aliases, then parse values using the answer type. Preserve distinctions that matter for the requested fact: currency, scale, sign, dates, fiscal periods, percent versus percentage points, and current versus list price. Exact extracted quantities use exact canonical equality; any rounding tolerance must follow the item's requested precision and be declared before scoring. Do not apply a global relative tolerance that can accept an incorrect large count or price. An answer containing incompatible alternatives is incorrect even if one matches the gold. Remaining semantic cases use a versioned grader rubric with an auditable human escalation path. Abstentions are incorrect, and grading explanations are retained.

Release and renewal. Near-duplicate images, paraphrased questions, and syndicated articles are grouped by event before release. Distinct URLs do not guarantee independent facts. Composition caps and deterministic release selection are versioned; the target count cannot override quality thresholds. Changed facts create new time-anchored items, while corrections create new versions with an explicit supersession record. A weekly source check may detect changes, but must not silently rewrite a frozen leaderboard's gold answers.

C Short-Circuit Certification and Cost Accounting

Order the $K = |\mathcal{M}|R$ closed-book and oracle trial outputs in advance. The exhaustive decision is the conjunction of K CB-failure indicators and K OR-success indicators. Short-circuit evaluation stops on the first false conjunct, then applies the same frozen release postprocessing to survivors.

Proposition. For fixed candidates, P0 outcomes, ordered trial outcomes, grader, and deterministic release postprocessing, exhaustive and short-circuit certification produce identical released item IDs.

Proof. If short-circuit evaluation rejects, it has observed a false conjunct in Equation 2, so exhaustive evaluation also rejects. If it accepts, all conjuncts have been evaluated and are true, so exhaustive evaluation accepts. Conversely, any exhaustive rejection contains a false conjunct; the short-circuit procedure encounters one no later than that position. Thus the automatically admitted sets agree. Identical deterministic postprocessing maps them to identical release sets. *QED*

This is a statement about evaluating recorded Boolean outcomes. Two new API runs can disagree because their outputs differ. An implementation audit therefore replays one cached prediction matrix, or uses genuinely replayable per-item trial randomness, and checks both the set hash and call ledger. Changing the order of globally consumed random numbers is not a replay.

Appropriate baselines. For n P0-passing candidates and n_{CB} candidates passing P1, unconditional exhaustive evaluation uses $2Kn$ predictions. A stronger baseline skips OR after CB rejection and uses $Kn + Kn_{\mathrm{CB}}$. Short-circuit cost is the number of actually reached trials; accepted items still use $2K = 24$. Report savings against both baselines, the per-model token ledger, API prices at execution time, generation/audit overhead, human time, wall-clock latency, total build cost, and cost per released item. Expected cost can be written without an independence assumption as

$$\mathrm{E}[\mathrm{Cost}] = \sum_j c_j \Pr(\text{trial } j \text{ is reached}), \tag{8}$$

for fixed per-trial costs c_j. Real token-dependent costs use the observed ledger. Proposal-side changes have different accepted sets and yields and belong in a construction tradeoff study, not an equal-output cost claim.

D Evaluation Records and Unresolved Evidence

An empirical evaluation manifest records model checkpoint/API revision, provider, date/time, prompts, supported decoding controls, image resolution, search provider and locale, page extraction, passage/context truncation, tool schemas, retries, maximum calls, token budgets, and timeouts. The reference design uses three fresh episodes per item and condition, temperature 0.2 and top-p 1 where supported, and a shared answer-format instruction. WS allows at most ten search/open actions and 16,000 visible text tokens, with at most 2,000 extracted tokens from a single page. The time limit is 120 seconds per episode. These are proposed protocol parameters; they are not parameters of a completed deployment.

Unsupported controls are disclosed rather than described as identical. The same checkpoint and common instruction do not make CB, WS, and OR computationally identical. CB/OR receive no external retrieval tools. Each condition has an independent context and item order is randomized. Live runs are interleaved over a narrow time window; archived replay is a separate condition because index drift can change available evidence.

Budget exhaustion and a submitted abstention count as operational failures. Infrastructure outages are logged separately; any retry rule is fixed before execution. Report completed-run coverage, operational accuracy on all scheduled runs, and accuracy conditional on a completed run. Do not silently drop failures. Retain the original first attempt when a retry is permitted. Report paired item-level differences and uncertainty clustered at the level of shared events/builds, with run variation kept inside each item.

Evidence labels. Let availability A be present, absent, or unresolved. Set $p_j = \Pr(A = j)$ and $u_j = \Pr(C = 1 \mid A = j)$ for each nonempty group. Then

$$\mathrm{Acc}_{\mathrm{WS}} = p_{\mathrm{present}} u_{\mathrm{present}} + p_{\mathrm{absent}} u_{\mathrm{absent}} + p_{\mathrm{unresolved}} u_{\mathrm{unresolved}}. \tag{9}$$

Do not replace the last group by absent. Treating observed labels as correct, the supporting-fact coverage lies between p_{present} and $p_{\mathrm{present}} + p_{\mathrm{unresolved}}$ until unresolved cases are resolved. False-positive and false-negative matcher errors can widen that uncertainty. Conditional rates on empty groups are undefined, not zero. The binary identity in the main text applies when all labels are resolved or explicitly to a reported resolved subset, with that subset's denominator.

The primary error table uses support-present, support-absent, and unresolved as a disjoint partition of incorrect WS answers. Tags for stale evidence, competing facts, or contradiction can overlap and use separate counts. An OR failure cannot assign a causal WS label, since the conditions contain different contexts. Mechanistic statements about noise require interventions.

E Empirical Studies Needed to Test the Joint Protocol

Visual contribution. Evaluate each frozen question with its original image, without an image, and with a matched distractor image. Add a text-only control that supplies the intended entity name but not the event answer. Retain question wording, search budget, and evaluation time. Audit distractor images to avoid answer leakage or unrelated changes in difficulty. Paired image effects measure the tested visual contribution; an average accuracy decrease does not prove image necessity for every item. Report per-category effects and the nonzero text-only residual, rather than only the aggregate drop.

Held-out certificate transfer. Use models outside the construction panel and, where possible, its model families. Record fresh $R = 4$ CB and OR trials under the same certification profile. Define $B = 1$ for no held-out CB success and $E = 1$ for all held-out OR successes. Report the complete 2×2 table of B, E, with joint transfer $\Pr(B = 1, E = 1)$. Marginal CB leakage and OR transfer do not determine this joint probability; they only give the bounds

$$\max(0, \Pr(B) + \Pr(E) - 1) \leq \Pr(B \wedge E) \leq \min(\Pr(B), \Pr(E)). \tag{10}$$

Any-of-twelve CB success is different from single-trial CB accuracy. Report fresh construction-panel runs separately from held-out runs and test ranking sensitivity after excluding overlapping families. Stability of a few ranks does not establish the absence of selection bias.

Joint filtering versus its components. To test the added value of the proposed combination, compare a common candidate pool under no P1/P2 filter, P1 alone, P2 alone, single-model joint filtering, and repeated multi-model joint filtering. Hold generation and P0 fixed, then report admitted counts, per-stage denominators, expert quality, held-out joint transfer, topic/answer-type shifts, and cost per release. Separately ablate P0, evidence-first generation, and constitution memory; these interventions change the candidate or accepted distribution. Inspect a probability sample of rejected items for false rejection. Quality among accepted items cannot establish rejection recall.

Comparable benchmark evaluation. Use the documented prior benchmarks without relabeling their original task definitions. Evaluate compatible subsets under a common model version, tool interface, budget, and grader, and report excluded task types. Compare distributions of residual no-search answerability, gold-context answerability where valid gold is available, and expert-validity criteria. Different source dates, task scopes, and search horizons prevent raw leaderboard scores from being interpreted as measurement superiority.

Controlled context and search interventions. For each item, construct clean gold, gold plus five irrelevant passages, and gold plus five conflicting passages. The two added-context conditions match actual token length, source-format cues, and gold position. Randomize and record passage order; verify that gold survives truncation and that conflicting passages refer to the intended perturbation (e.g., an earlier price or another fiscal period). Human reviewers validate the perturbation. The clean baseline is intentionally shorter. Report both clean-to-irrelevant and irrelevant-to-conflicting contrasts with paired transitions.

For retrieval depths 3, 10, and 20, specify whether the intervention changes search results fetched, passages opened, or passages exposed to the model. Use a common retrieval snapshot to isolate depth, and a separately reported live rerun to measure operational behavior. A top-20-to-five reranker sees only allowed query/image and candidate content, never gold answers or certificates. Record reranker identity, cost, gold retention, context length, and WS accuracy. A gain in accuracy may coexist with reduced coverage; the tradeoff must be reported. None of these tests alone identifies a scaling law.

Rolling validity. Report actual daily timestamps, released counts and shortfalls, unique events, duplicate removals, source-domain concentration, answer types, and per-day outcomes. Use an event-disjoint prospective period to test changes to the generator or constitution. If an event spans several days, group it across those days or use an appropriate block analysis; treating all daily items as independent understates uncertainty. The synthetic fixture below contains no events spanning multiple builds.

F Reproducible Synthetic Study

The demonstration contains numerical variables only: no natural-language questions, images, fetched pages, expert labels, or actual model outputs are generated. This distinction is necessary because numerical consistency cannot validate a benchmark's selection distribution or source correctness.

F.1 Data-generating assumptions

For item i in build b and event e, draw independent latent terms $d_b \sim \mathcal{N}(0, 0.40^2)$, $d_e \sim \mathcal{N}(0, 0.50^2)$, and $d_i \sim \mathcal{N}(0, 0.65^2)$, shared across configurations. Familiarity f_i, retrieval r_i, reading s_i, ambiguity a_i, position z_i, and context alignment h_i are standard normal. Each item receives one of six uniform topic IDs; the configuration–topic loading t_{mi} is drawn from $\mathcal{N}(0, 0.11^2)$ and retained in the metadata. Writing σ for the logistic function, baseline probabilities are

$$g_{mi} = \sigma(\alpha_m - .70d_b - .45d_e - .35d_i + .25r_i + t_{mi}), \tag{11}$$

$$v_{mi} = \sigma(\beta_m - .35d_b - .45d_e - .55d_i + .25s_i - .12a_i - .5t_{mi}), \tag{12}$$

$$w_{mi} = \sigma(\gamma_m - .25d_b - .20d_e - .25d_i + .18f_i), \tag{13}$$

$$b_{mi} = \sigma(\delta_m - .25d_b - .20d_e - .25d_i + .55f_i), \tag{14}$$

$$o_{mi} = \sigma(\omega_m - .25d_b - .25d_e - .45d_i + .35s_i - .4t_{mi}). \tag{15}$$

For each item, independent uniforms for CB, availability, conditional correctness with/without support, and OR are shared across configurations. Set $G_{mi} = 1[u_i^G < g_{mi}]$ and use v_{mi} for WS correctness when $G_{mi} = 1$, otherwise w_{mi}. CB and OR use b_{mi} and o_{mi}. Shared uniforms produce a numerical pairing, not an assumption that real agents have shared random outputs. The configuration parameters and every latent/uniform value are included in the released files.

F.2 Intervention equations

At retrieval depth k, shift the retrieval logit by $.48 \log(k/10)$ and the conditional-reading logit by $(\lambda_m - .04a_i) \log(k/10)$. Using the same uniforms makes evidence acquisition nested across depths, while reading can degrade. Reranking retains each depth-20 supporting fact with probability $\sigma(3 - .12a_i + .10s_i)$ using a separate shared uniform; its reading logit is the baseline reading logit plus $\tau_m + .10h_i$, with configuration-specific τ_m retained in the code. There is no gold-informed real reranker in this fixture.

The irrelevant-context shift to the OR logit is $-\xi_m + .16h_i + .10z_i$. The conflicting-context shift is $-\zeta_m(.85 + .20\sigma(a_i)) + .10h_i - .18z_i$. Both have six abstract context units, versus one for clean OR. These are numerical controls, not verified token-matched passages. A real test must implement the passage construction in Appendix E.

F.3 Outputs, pairing, and verification

The fixture contains 72,000 baseline item–configuration rows and 168,000 intervention rows. Each aggregate is computed from those records; the four joint cells sum to 6,000 for every configuration. Bootstrap resampling draws 30 whole builds with replacement 2,000 times, using the same build weights for every configuration and condition. Conditional rates use resampled count ratios, not means of daily ratios. Paired changes retain both wrong-to-correct and correct-to-wrong transitions.

The random generator is NumPy PCG64 with root seed 20260905 and separate SeedSequence streams for fixture generation and bootstrap resampling. The design was fixed internally before

Config.	Irrel. vs clean	Conflict vs irrel.	$k = 20$ vs 3	Rerank vs 20
A01	-0.5 [-0.7, -0.3]	-10.1 [-10.8, -9.3]	+11.0 [+10.1, +11.9]	-1.7 [-2.3, -1.2]
A04	-0.9 [-1.1, -0.8]	-6.9 [-7.6, -6.3]	+11.8 [+10.9, +12.8]	-2.0 [-2.5, -1.6]
A07	-1.3 [-1.6, -1.0]	-14.0 [-14.7, -13.2]	-1.4 [-2.4, -0.2]	+7.5 [+6.7, +8.3]
A10	-0.2 [-0.5, -0.0]	-13.4 [-14.2, -12.6]	+6.7 [+5.6, +7.7]	+4.1 [+3.5, +4.8]

Table 5: **Synthetic protocol demonstration: paired changes.** Entries are percentage-point changes [95% paired build-bootstrap intervals]. The first two columns change oracle context; the latter two change with-search retrieval. Irrelevant and conflicting contexts have matched abstract context units, without generated passages. All contrasts share the same 6,000 synthetic items. These numerical perturbations do not measure causal effects in real systems.

generation, not externally preregistered. The saved design, parameters, and script disclose that the context effects are chosen assumptions. No seed search or parameter fitting to target results was performed. The script records the NumPy version and compressed raw-data SHA-256 hashes; gzip timestamps are fixed for stable raw-file hashes. Figure files need not be byte-identical because rendering metadata can differ.

An independent CSV reader checks unique item keys, shared item sets, per-build counts, all joint cells, every reported baseline aggregate, paired intervention counts, and the decomposition identity. These checks validate the numerical artifact, not the empirical assumptions. The implementation and verification commands are:

```
python simulation/generate_simulation.py
python simulation/validate_from_csv.py
python simulation/audit_sampling_design.py
```

The generator requires NumPy, SciPy, and Matplotlib. The CSV validator uses Python's standard library; the analytical audit script requires SciPy.

G Complete Evaluation Results and Paired Comparisons

This appendix retains the complete configuration order used in the main figures. The baseline has 6,000 unique numerical items shared by all twelve configurations; the resulting 72,000 rows are not 72,000 independent items. The main table reports conditional denominators through coverage and the full joint counts immediately below. Every marginal rate, conditional rate, and paired contrast can be reconstructed from the delivered records.

Config.	CB	WS [95% interval]	OR	Cov	U_1	U_0
A01	9.5	61.9 [59.9, 63.7]	94.4	74.0	80.9	7.6
A02	10.0	60.6 [58.5, 62.6]	95.0	70.0	82.6	9.3
A03	8.4	57.6 [55.7, 59.4]	94.1	74.6	75.1	6.3
A04	10.0	58.1 [55.9, 60.1]	93.9	66.9	81.9	9.7
A05	8.9	53.8 [51.9, 55.5]	93.5	71.9	71.9	7.5
A06	11.2	53.8 [51.6, 56.0]	94.2	63.1	78.9	10.9
A07	8.7	50.7 [48.9, 52.5]	93.5	73.1	66.5	7.9
A08	10.9	50.9 [48.9, 52.9]	92.7	66.1	71.9	10.2
A09	9.5	46.7 [44.8, 48.6]	92.7	67.1	65.5	8.3
A10	10.4	47.1 [45.1, 49.0]	92.5	60.4	70.8	10.8
A11	9.6	43.6 [41.6, 45.6]	91.8	67.8	59.9	9.4
A12	11.7	44.5 [42.4, 46.5]	92.0	62.3	64.1	12.3

Table 6: **Synthetic protocol demonstration.** Anonymous numerical configurations on 30 simulated builds of 200 items each. All rates are percentages. Intervals use 2,000 paired build-cluster bootstrap resamples and quantify variation in the synthetic generator only. U_1 and U_0 use the observed joint-cell denominators. These are not measurements of real models.

Config.	$C=1, G=1$	$C=1, G=0$	$C=0, G=1$	$C=0, G=0$	$P(C=1, G=0)$
A01	3,594	118	847	1,441	1.97
A02	3,468	167	729	1,636	2.78
A03	3,360	96	1,117	1,427	1.60
A04	3,291	192	725	1,792	3.20
A05	3,100	126	1,213	1,561	2.10
A06	2,989	241	800	1,970	4.02
A07	2,916	128	1,468	1,488	2.13
A08	2,850	207	1,114	1,829	3.45
A09	2,639	163	1,388	1,810	2.72
A10	2,568	257	1,057	2,118	4.28
A11	2,437	181	1,630	1,752	3.02
A12	2,396	277	1,343	1,984	4.62

Table 7: **All synthetic joint counts.** Each row totals 6,000. The last column is a percentage of all items and equals the contribution $(1 - \mathrm{Cov})U_0$ to WS; it is not the conditional rate U_0.

Intervals and near-ties. Whole-build resampling preserves all configurations, conditions, and event clusters inside each selected build. It is appropriate for the assumed fixture, whose events do not recur across builds. A deployment with events spanning days would require a different cluster definition. Intervals in Table 8 are for paired differences and do not follow from subtracting endpoints of marginal intervals. The six adjacent contrasts are descriptive summaries in the fixed configuration order; they are not a confirmatory family of tests with controlled family-wise error. A01–A02 and A11–A12 have intervals excluding zero in this fixture, while the four middle contrasts cross zero. This pattern is compatible with overlapping individual rate intervals because the records are paired.

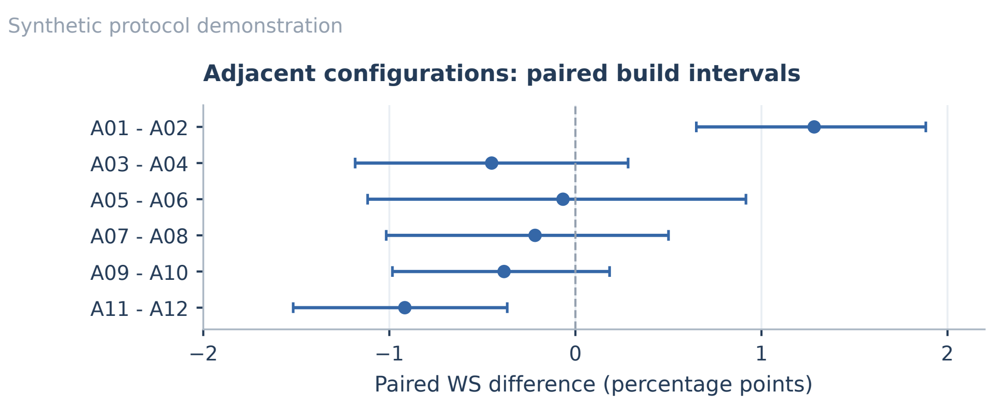

Figure 8: **Paired differences preserve the shared-item comparison.** Point estimates and 95% build intervals use the baseline bootstrap stream. The synthetic configurations are retained in their original order. These intervals express numerical sampling variation, without claims about differences between existing models.

Rolling analysis. The heatmap in Figure 3 subtracts each configuration's aggregate WS score from each of its 30 build scores. This centers rows without changing their order or their absolute scores in Table 6. The common diverging scale prevents independent color normalization from exaggerating stability or variation. `analysis_v3/daily_evaluation.csv` retains all 360 build-level scores, coverages, denominators, and deviations. These are repeated draws from a fixed generator, not calendar evidence of model improvement or benchmark renewal. They do not measure real source turnover or duplication.

Contrast	Δ WS [95% interval]	A only	B only	Both correct	Both wrong
A01 minus A02	1.28 [0.65, 1.88]	175	98	3537	2190
A03 minus A04	-0.45 [-1.18, 0.28]	310	337	3146	2207
A05 minus A06	-0.07 [-1.12, 0.92]	325	329	2901	2445
A07 minus A08	-0.22 [-1.02, 0.50]	236	249	2808	2707
A09 minus A10	-0.38 [-0.98, 0.18]	226	249	2576	2949
A11 minus A12	-0.92 [-1.52, -0.37]	168	223	2450	3159

Table 8: **Descriptive paired comparisons.** A and B denote the first and second named configurations. Four transition cells sum to 6,000. Intervals resample complete builds and are pointwise, without multiplicity adjustment.

Intervention transitions. The four cells for each before/after correctness pair are both correct, both wrong, corrected, and newly wrong. The net accuracy change is $(n_{\text{corrected}} - n_{\text{newly wrong}})/N$. This accounting distinguishes a small net effect caused by many opposing transitions from one caused by almost no changes. In the A01 reranking example, 80 corrected and 183 newly wrong responses yield $100(-103/6000) = -1.7167$ pp. No transition is discarded on the basis of whether it supports an improvement narrative. The saved paired-intervention JSON contains this partition for every reported contrast, including matched context and retrieval depth.

H Visual-Control Design and Complete Results

The visual-control study is a numerical extension of the existing fixture, not a second generated baseline. The original probabilities, uniforms, and full-image outcomes are immutable inputs. All conditions use the same 6,000 items and A01–A12 configurations, producing 288,000 rows across four conditions. The control seed is 2026090505, with separate streams for donor links and 2,000 bootstrap draws. Because this bootstrap stream differs from the baseline stream, marginal intervals can have small Monte Carlo differences even when the underlying full-image estimate is identical. Within each analysis, all contrasts share the same weights.

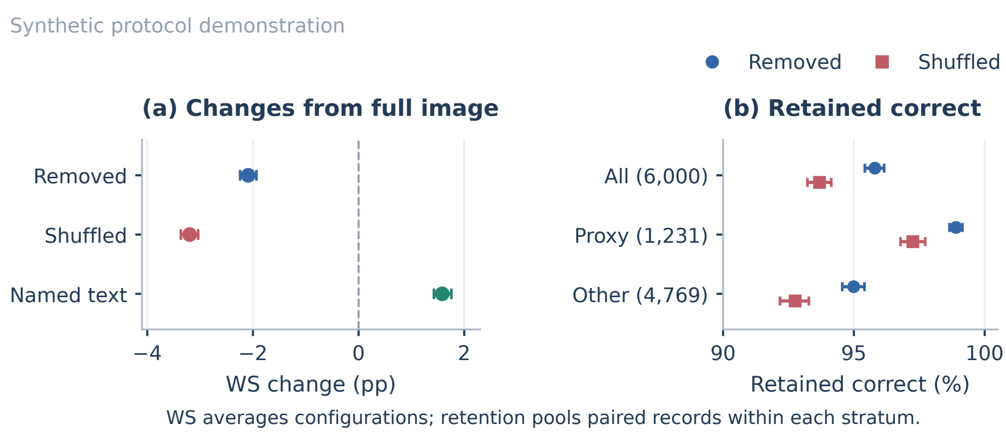

Figure 9: **A visual-control drop is not an item-necessity certificate.** (a) Paired changes relative to the unchanged full-image numerical baseline. (b) The fraction of originally correct outputs retained after each change, pooling configuration–item records. Counts name unique items; the proxy strata are latent-defined assumptions. Pointwise build intervals retain pairing across conditions. All inputs are numerical surrogates.

H.1 Explicit probability interventions

Let $S_i = 1[f_i \geq 0.8]$ be the familiarity-based shortcut proxy, and $d_i^V = 0.70 + 0.45\sigma(a_i)$. This proxy selects 1,231 items; its label does not claim an observed linguistic shortcut. Each configuration has query sensitivity q_m and reading sensitivity r_m, in fixed order:

$$(q_m)_{1:12} = (.18, .09, .30, .04, .22, .12, .35, .16, .27, .06, .24, .14),$$
$$(r_m)_{1:12} = (.025, .055, .010, .035, .050, .000, .015, .045, .030, .065, .020, .040).$$

For image removal, shift the baseline retrieval and reading logits by

$$\Delta_G = -q_m d_i^V(1 - .80S_i) + .035\tanh(z_i),$$
$$\Delta_{C1} = -r_m d_i^V(1 - .70S_i) + .020\tanh(h_i),$$
$$\Delta_{C0} = -.040(1 - S_i) + .015S_i.$$

For the named-entity text control, the respective shifts are $.080 + .40q_m d_i^V(1 - .65S_i) + .030\tanh(h_i)$, $-.20r_m d_i^V + .010\tanh(h_i)$, and $.030(1 - .50S_i)$. Thus the generator explicitly assumes that naming the referent can help retrieval. The result does not independently validate this assumption.

Each shuffled donor is chosen through a fixed within-build bijection: an event offset from 1 to 19 and an item offset from 0 to 9 map each target to a different event. Donors never equal their targets. The shuffled retrieval logit shift is $-(q_m + .070)d_i^V(1 - .60S_i) + .070\tanh(r_i^{\text{donor}} - r_i)$; the with-support shift is $-(r_m + .025)d_i^V(1 - .50S_i) + .025\tanh(s_i^{\text{donor}} - s_i)$; the without-support shift is $-.060(1 - .50S_i) + .015\tanh(f_i^{\text{donor}} - f_i)$. Here r_i is the item retrieval latent and r_m the configuration sensitivity; their subscripts distinguish them. The implementation retains all donor links, probabilities, and shifted logits.

For each condition, transform shifted logits by σ and threshold against the original availability and conditional-correctness uniforms. This yields paired changes, including losses and occasional gains. It does not render, remove, or replace actual images. The source design and equations are saved in `analysis_v3/validity_DESIGN.md`.

H.2 Retained correctness and stratification

Condition	WS [95% interval]	Δ WS [95% interval]	Retained	Ratio
Full image	52.45 [50.48, 54.36]	0.00 [0.00, 0.00]	100.00	100.00
Removed	50.36 [48.36, 52.32]	-2.09 [-2.24, -1.93]	95.80	96.02
Shuffled	49.25 [47.24, 51.23]	-3.20 [-3.36, -3.04]	93.69	93.90
Named text	54.03 [52.15, 55.84]	1.58 [1.43, 1.76]	99.70	103.02

Table 9: **Aggregates across the 12 configurations.** WS averages equal-sized configuration samples; conditional metrics pool their underlying counts. WS, retention, and accuracy ratio are percentages; Δ is in percentage points. The ratio can exceed 100%, unlike conditional retention.

The retained-correct rate is $\Pr(C_{\text{control}} = 1 \mid C_{\text{full}} = 1)$. It differs from the ratio $\Pr(C_{\text{control}} = 1)/\Pr(C_{\text{full}} = 1)$ because previously incorrect responses can become correct. Table 9 reports both: the named-text ratio exceeds 100% while retention does not. WS and coverage pool equal-sized configuration samples, equivalently averaging the twelve configuration scores. Conditional metrics instead pool the underlying configuration–item counts and use their actual denominators. The accuracy ratio divides the two pooled WS rates. Each bootstrap draw recomputes these same estimands from resampled counts.

The proxy stratum attenuates image-removal effects by construction. The observed difference between strata therefore illustrates sensitivity to an assumption, not a discovered prevalence or mechanism. Its denominator is 1,231 unique items; the complementary stratum contains 4,769. The raw paired records retain every correctness/availability cell in each condition. The summary file additionally reports coverage, U_1, U_0, gains among baseline errors, and all correctness and coverage transitions for every configuration, condition, and stratum.

An empirical visual study should use actual controlled images and inspect textual shortcuts, ambiguity, and answer leakage. These four aggregate controls do not separately identify query-side and answer-side pathways: the same image may affect both query formation and reading. A factorial scaffold that fixes retrieved context while varying only the answering input can separate these pathways, with search-query effects analyzed in a complementary condition. Expert judgments remain necessary to decide whether a particular item satisfies the intended visual-reference contract.

Config.	Removed Δ WS	Shuffled Δ WS	Named text Δ WS
A01	-2.08 [-2.52, -1.67]	-3.05 [-3.53, -2.52]	1.63 [1.32, 1.95]
A02	-1.60 [-1.93, -1.27]	-2.58 [-2.98, -2.17]	1.25 [0.87, 1.67]
A03	-2.75 [-3.15, -2.32]	-4.12 [-4.60, -3.63]	1.73 [1.40, 2.10]
A04	-0.63 [-0.87, -0.40]	-1.65 [-2.02, -1.32]	1.48 [1.20, 1.78]
A05	-2.55 [-2.98, -2.13]	-3.65 [-4.03, -3.25]	1.43 [1.15, 1.72]
A06	-1.68 [-2.05, -1.37]	-2.92 [-3.30, -2.58]	1.98 [1.55, 2.40]
A07	-2.95 [-3.30, -2.58]	-4.05 [-4.42, -3.68]	1.78 [1.43, 2.15]
A08	-2.37 [-2.82, -1.93]	-3.52 [-4.02, -3.05]	1.58 [1.23, 1.95]
A09	-2.97 [-3.47, -2.50]	-4.00 [-4.60, -3.43]	1.83 [1.47, 2.18]
A10	-1.20 [-1.52, -0.90]	-2.65 [-3.03, -2.25]	1.37 [1.12, 1.67]
A11	-2.17 [-2.53, -1.78]	-3.15 [-3.52, -2.75]	1.77 [1.35, 2.25]
A12	-2.12 [-2.42, -1.83]	-3.03 [-3.33, -2.75]	1.15 [0.80, 1.50]

Table 10: **All visual-control effects in the numerical fixture.** Values are paired changes from the unchanged full-image baseline in percentage points, with pointwise 95% build intervals. Each cell uses 6,000 shared items.

Stratum	Unique n	Removed Δ WS (pp)	Retained correct (%)
All	6,000	-2.09 [-2.24, -1.93]	95.80 [95.42, 96.16]
Proxy	1,231	-0.43 [-0.57, -0.30]	98.90 [98.65, 99.15]
Other	4,769	-2.52 [-2.69, -2.33]	94.99 [94.56, 95.40]

Table 11: **Latent-defined strata under image removal.** Proxy means familiarity ≥ 0.8. Stratum membership and attenuation are generator assumptions, not observed shortcut annotations. Intervals recompute each conditional denominator on resampled builds.

I Construction Selection, Transfer, and Cost Replay

I.1 A separate candidate population

This extension uses seed 2026090506 and 30 builds of 600 candidates, each with 60 event clusters of ten candidates. It does not select items from the 6,000-item evaluation fixture. The generator assigns build, event, candidate familiarity, candidate clarity, and family effects. Construction responders C1/C2 share one synthetic family and C3 another; held-out responders H1/H2 and H3 use two different synthetic families. Both panels share candidate, event, and build terms, so family separation does not imply statistical independence.

The CB logit has intercept -4.1, coefficients .50 for the build term, .45 for the event term, 1.00 for familiarity, .15 for inverse clarity, and .45 for its family effect. The OR logit has intercept 3.3 and coefficients .35 for build, .35 for event, .90 for clarity, $-.10$ for familiarity, and .40 for family. Configuration offsets and latent distributions are fixed in `construction_DESIGN.md`. Within each family, CB and OR effects have correlation .25. A separate P0 Bernoulli gate depends on clarity and a visual latent. P0 is a recorded numerical gate, not an independently validated quality label.

All 48 Boolean panel outcomes per candidate are generated before applying any procedure: three responders times four trials times two conditions times two panels, for 864,000 outcomes in total. Trials share latent probabilities while using separately drawn uniforms. The reference CB/OR trial order is fixed responder-major, with CB evaluated before OR. This cached-outcome construction makes execution comparisons exact and auditable. It does not assume that repeated real API calls are deterministic.

I.2 Selection policies and both cohort denominators

Each policy in Table 12 starts from the same 15,459 P0-eligible candidates. P1-only and P2-only add a single predicate; the one-responder joint policy uses the prespecified C1's four trials per condition. The full policy requires all twelve CB failures and twelve OR successes. The admitted populations

Policy	Admitted n	CB any (%)	OR all (%)	Joint pass [95% interval]
P0	15,459	26.86	61.26	45.48 [42.85, 48.15]
P0 + P1	11,179	19.65	62.14	50.44 [48.10, 52.75]
P0 + P2	9,560	25.41	70.47	52.92 [50.62, 55.35]
P0 + one joint	11,390	23.42	65.29	50.50 [48.13, 52.88]
P0 + full joint	7,004	18.48	71.17	58.24 [56.21, 60.23]

Table 12: **Synthetic selection on a common P0 pool.** Each row uses its own admitted denominator. CB any is at least one of 12 held-out successes; OR all is 12 of 12 successes. Joint pass requires zero CB successes and all OR successes. The single-configuration policy uses fixed C1.

therefore differ. Their held-out quality rates describe selection; comparing them cannot show that a filter improves a fixed model on a fixed item set.

Policy	$B{=}1, E{=}1$	$B{=}1, E{=}0$	$B{=}0, E{=}1$	$B{=}0, E{=}0$	Total n
P0	7,031	4,275	2,439	1,714	15,459
P0 + P1	5,639	3,343	1,308	889	11,179
P0 + P2	5,059	2,072	1,678	751	9,560
P0 + one joint	5,752	2,971	1,685	982	11,390
P0 + full joint	4,079	1,631	906	388	7,004

Table 13: **Complete held-out joint cells: admitted cohort.** $B = 1$ denotes all CB trials failing; $E = 1$ denotes all OR trials succeeding. These are joint counts, not products of marginal rates.

Policy	$B{=}1, E{=}1$	$B{=}1, E{=}0$	$B{=}0, E{=}1$	$B{=}0, E{=}0$	Total n
P0	2,724	1,691	936	649	6,000
P0 + P1	2,956	1,824	727	493	6,000
P0 + P2	3,126	1,332	1,055	487	6,000
P0 + one joint	2,964	1,619	876	541	6,000
P0 + full joint	3,320	1,362	771	333	5,786

Table 14: **Complete held-out joint cells: released cohort.** $B = 1$ denotes all CB trials failing; $E = 1$ denotes all OR trials succeeding. These are joint counts, not products of marginal rates.

The full-joint admitted cohort contains 4,079 items passing both held-out predicates, 1,631 passing only P1, 906 passing only P2, and 388 passing neither. Applying the common per-build priority and cap changes these counts to 3,320, 1,362, 771, and 333. Separate tables preserve the distinction between the 7,004 admitted and 5,786 released denominators. A marginal CB failure rate and a marginal OR success rate cannot reconstruct these cells.

Every build processes all 600 candidates and uses an independent random priority drawn once and shared by all policies. It releases up to the first 200 admitted candidates. Daily full-joint admissions range from 143 to 310, with median 241. Five builds contribute 214 missing target slots; 1,218 admissions exceed other builds' caps. Surpluses are not borrowed across builds. No new outcome is generated to fill a quota. This stream has no recurring events across days, source outages, or generator updates; an operational renewal study must measure those factors.

I.3 Execution equivalence and accounting limits

Exhaustive evaluation checks all 24 predictions for every P0 survivor. Block skipping checks all twelve CB predictions but skips the entire OR block on a P1 failure. Exact short-circuit evaluation stops each conjunction at its first failing literal. The latter avoids 105,820 predictions relative to exhaustive evaluation and 54,460 relative to block skipping. Every admitted candidate nevertheless retains all 24 checked predictions. The accepted-ID and released-ID sets are equal under all three procedures.

Procedure	Checked predictions	Per admitted item	Per released item
Exhaustive	371,016	52.97	64.12
Block skip	319,656	45.64	55.25
Short circuit	265,196	37.86	45.83

Table 15: **Execution counts on the same cached outcomes.** All procedures admit 7,004 and release 5,786 identical item IDs. Counts include trials spent on rejected P0-eligible candidates, but exclude generation, P0, held-out evaluation, and human review.

The reported saving is a ratio of total avoided predictions to the relevant total baseline count, recomputed with the same build multiplicities in each bootstrap draw. It is not an average of daily saving percentages. The intervals are 28.52% $[26.31, 30.69]$ and 17.04% $[15.87, 18.30]$ against the two respective baselines. Equal unit costs exclude differences in token lengths, API pricing, parallelism, failures, and latency. Generation, P0, held-out checks, and human review are outside this execution comparison and must be included in a deployment budget.

I.4 Reproduction and independent checks

The `analysis_v3/` directory stores fixed designs, seeds, original CSV records, shared bootstrap weights, summaries, and validators. `construction_validate.py` independently recomputes eligibility, all five policy cohorts, every held-out joint cell, day-level shortfalls, procedure counts, exact ID equality, and 2,000 bootstrap intervals without importing the generator. The visual-control validator independently checks input hashes, donor bijections, probability replay, unchanged baselines, all summary rates, transitions, and intervals. Analytical audit detection reduces to the original hypergeometric formula when $s = 1$.

These independent checks verify numerical reproducibility and fixed-input execution equivalence. Empirical validity, reviewer reliability, and deployed costs require the separate measurements specified in the protocol.